

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

7th Semester of B. Pharm. Examination

University Theory Examination Oct/Nov 2017

PH404 Pharmaceutical Analysis III

Date: 03.10.17, Tuesday

Time: 1:30p.m to 4:30p.m

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q 1 Attempt any TWO of the following

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|---|--|----|
| A | (i) Enlist various factors which may affect HPTLC separation. | 03 |
| | (ii) Write a note on different steps involved in HPTLC. | 07 |
| B | Explain the following terms and their significance: Retention time, tailing factor, theoretical plate, guard column, gradient elution. | 10 |
| C | Write short note: | |
| | (i) Ion-exchange chromatography | 05 |
| | (ii) Super critical fluid chromatography | 05 |

Q 2 Attempt any TWO of the following

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|---|--|----|
| A | (i) Draw the schematic diagram of HPLC | 03 |
| | (ii) Enlist the different types of detectors used in HPLC | 02 |
| | (iii) Give the classification of Chromatographic techniques. | 05 |
| B | (i) Give an account on the types of columns used in gas chromatography. | 06 |
| | (ii) State the ideal requirements of carrier gas used in gas chromatography. | 04 |
| C | Describe the construction and working of the following gas chromatographic detectors | 10 |
| | (i) Electron Capture Detector | |
| | (ii) Flame Ionisation Detector | |

SECTION- II

Q 3 Attempt any TWO of the following

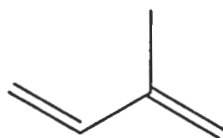
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|---|---|----|
| A | Explain the principle of FT-IR spectroscopy along with suitable schematic diagram of the instrumentation. State its advantage over the Dispersive IR spectroscopic technique. | 10 |
|---|---|----|

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|---|-------|---|----|
| B | (i) | Explain the term Fermi Resonance. | 02 |
| | (ii) | Justify: Different types of amines show different peaks in the functional group region of the IR Spectra. | 02 |
| | (iii) | Explain ATR with a suitable diagram. | 06 |
| C | (i) | Explain the phenomenon of Fluorescence and Phosphorescence with the Jablonski Diagram. | 05 |
| | (ii) | Define Quenching and explain the different types of quenching. | 05 |

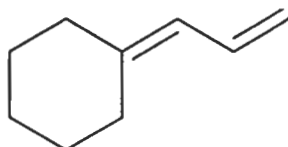
Q 4 Attempt any TWO of the following

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|---|-------|---|----|
| A | (i) | State Beer Lamberts law and explain the reasons for the deviations from the law. | 05 |
| | (ii) | Define (a) Chromophore, (b) Auxochrome, (c) specific absorptivity, (d) Bathochromic shift, (d) λ_{max} | 05 |
| B | | Explain the factors affecting absorption of UV radiation by molecules. | 10 |
| C | (i) | Justify: Quartz material is used for the construction of cuvette used in the UV Spectroscopy analysis. | 02 |
| | (ii) | PDA detector is preferred over other UV detectors. | 02 |
| | (iii) | Calculate the Lambda max for the following structures | 04 |

(a)



(b)



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|------|---|----|
| (iv) | Draw a schematic diagram of double beam UV Spectrophotometer. | 02 |
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